

# Executive Summary

Effective inventory planning is critical for retail businesses because both overstocking and stock shortages directly impact profitability and customer satisfaction. The objective of this project was to develop an intelligent sales forecasting system capable of predicting future product demand, identifying unusual sales patterns, and segmenting products based on demand behavior.

Historical retail sales data spanning four years was analyzed to understand sales trends, seasonality, and regional performance. Three forecasting techniques, namely SARIMA, Prophet, and XGBoost, were developed and evaluated using industry-standard performance metrics. Among these models, **XGBoost achieved the lowest Mean Absolute Percentage Error (MAPE), making it the most accurate forecasting model for this dataset.** In addition to forecasting, anomaly detection techniques identified unusual sales periods, while K-Means clustering grouped product categories with similar demand characteristics to support inventory planning.

The developed system demonstrates how historical sales data can be transformed into actionable business insights that support better purchasing decisions, improve inventory utilization, and reduce operational costs.

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## Business Objectives

The project was designed to achieve the following objectives:

- Forecast future monthly sales with high accuracy.
- Identify seasonal demand patterns.
- Detect unusual increases or decreases in sales.
- Group products with similar demand characteristics.
- Support inventory planning using data-driven insights.
- Build an interactive dashboard for business users.

## Data Analysis Highlights

The exploratory analysis revealed several important business patterns.

Technology products consistently generated the highest sales revenue, indicating that they remain one of the strongest contributors to overall business performance. Sales exhibited clear

seasonal fluctuations throughout the year, suggesting that customer purchasing behavior follows recurring annual trends. Regional analysis also showed noticeable differences in sales performance, emphasizing the importance of localized inventory planning rather than maintaining identical stock levels across all regions.

Time-series decomposition confirmed the presence of long-term growth along with repeating seasonal behavior, validating the suitability of forecasting models for future demand prediction.

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## Forecasting Model Comparison

Three forecasting models were evaluated.

| Model          | Purpose                          | Observation                                                                                                        |
|----------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| SARIMA         | Statistical Time Series Model    | Successfully captured long-term trend and seasonality.                                                             |
| Prophet        | Business Time Series Forecasting | Produced reliable forecasts while clearly modeling yearly seasonal effects.                                        |
| <b>XGBoost</b> | Machine Learning Forecasting     | <b>Delivered the lowest MAPE and the highest forecasting accuracy, making it the recommended production model.</b> |

Although Prophet provided excellent visualization of long-term seasonal patterns, **XGBoost achieved superior forecasting accuracy based on quantitative evaluation metrics and is therefore recommended for operational deployment.**

## Anomaly Detection

Two anomaly detection approaches were implemented:

- Isolation Forest
- Z-Score Detection

Both methods successfully identified weeks where sales deviated significantly from normal business patterns. These anomalies likely correspond to promotional campaigns, holiday shopping periods, or unexpected market events. Monitoring such events enables businesses to react quickly by adjusting inventory allocation and replenishment strategies.

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# Product Demand Segmentation

K-Means clustering grouped product sub-categories into four demand segments based on their sales characteristics.

The resulting clusters represent products with similar demand behavior, allowing inventory managers to adopt different stocking strategies rather than treating all products equally. High-demand products should receive priority during replenishment, while lower-demand products can be stocked more conservatively to minimize excess inventory costs.

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## Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Deploy the XGBoost forecasting model for monthly inventory planning because it achieved the highest forecasting accuracy.
2. Increase inventory allocation before forecasted seasonal demand peaks to minimize stock shortages and lost sales.
3. Continuously monitor anomaly reports to quickly identify unexpected changes in customer demand and adjust procurement strategies accordingly.

## Business Impact

If implemented in a retail environment, the proposed system can provide several measurable benefits:

- Improved inventory utilization.
  - Reduced stock shortages.
  - Lower inventory holding costs.
  - Better demand planning.
  - Faster business decision-making.
  - Increased customer satisfaction through improved product availability.
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# Project Limitations

The forecasting models rely exclusively on historical sales records and do not incorporate external influences such as marketing campaigns, holidays, competitor activities, economic conditions, or weather. Including these additional variables could further improve forecasting accuracy in future versions of the system.

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## Conclusion

This project successfully demonstrates the practical application of time-series forecasting, anomaly detection, clustering, and business analytics within a retail environment. By combining statistical methods with machine learning, the system delivers reliable demand forecasts and actionable insights that support smarter inventory management decisions. Among the evaluated forecasting models, **XGBoost emerged as the preferred production model due to its superior predictive accuracy**, while the overall analytical framework provides a scalable foundation for future retail decision-support systems.